Prenatal Participation in WIC and AFDC Programs: Associations with Children's Reading Achievement

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Introduction

Early Childhood is the most valuable development phase, and during this time, family resources and prenatal conditions are the ones significantly influential on lasting cognitive and educational outcomes. It is proven over time that children, even after five years old, experience prolonged effects on their academic performance if they are not adequately nourished or deprived of enough family income, as established by research (Birch & Gussow, 1996; Brooks-Gunn & Duncan, 1997). In this regard, the U.S. government has two possibly vital programs - the Women, Infants, and Children (WIC) Nutritional Program and the Aid to Families with Dependent Children (AFDC) Program, which are meant to assist low-income pregnant mothers with dietary and financial help.

This research employs the Child Development Supplement of the Panel Study of Income Dynamics (PSID-CDS) as the data sources to make a deep investigation on whether prenatal participation in WIC and AFDC affects children's later reading achievement. It involves an exploration of the question in holistically applying such program participation in predicting reading scores, after accounting for child age, family income, low-birth-weight status, and the quality of the home environment. It is possible to derive the following objectives: estimate the independent effect of each variable, variance explained, and the exclusive contribution of each program. Although the correlation rather than the cause of the analyzed nutritional and income support outcome was examined in the research, the results contributed to the important policy information regarding the impact of early childhood nutrition and income support on later educational achievement.

Methods

Data Source and Sample

The data came from the Child Development Supplement (CDS) of the Panel Study of Income Dynamics (PSID), a US nationally representative longitudinal survey. The analytical sample contains 1,845 children between the ages of 3 and 13 in 1997, with complete information on all variables studied. PSID-CDS contains ample data about children's family background, home environment, and developmental outcomes, which makes it possible to draw population-level conclusions regarding early childhood conditions and academic achievement.

Measures

The dependent variable is Woodcock–Johnson Revised Reading Achievement Age Standardized Score (readss97), which has a continuous scale with values ranging from 47.5 to 165.5, and it is used to indicate children's reading abilities as a function of age norms.

The key independent variables are both the participation of pregnant in the women, infants, and Children Nutrition Program (WICpreg) and the financial aid program offered to families with children (AFDCpreg); both are dummy variables where 0 means No and 1 means Yes.

Control variables include the child's age in 1997 (AGE97, in years), total family income in 1997 (faminc97, USD converted in 2002 constant dollars), and low birth weight (bthw, 0 non-low birth weight, 1 low birth weight), along with home environment quality (HOME97, composite, score range 7-27, emotional and cognitive stimulation at home; Bradley et al., 1988).

Analytic Strategy

To estimate the association between prenatal program participation and child reading achievement, a series of nested ordinary least squares (OLS) regression models were fitted.

- Model 0 (M0) included only control variables (AGE97, faminc97, bthwht, HOME97).
- Model 1 (M1) added WIC participation.
- Model 2 (M2) added AFDC participation.
- Model 3 (M3) included both policy variables simultaneously.

The relevance of R^2 for model comparisons is that it shows the amount of variance accounted for by the models in total, while changes in $R^2(\Delta R^2)$ demonstrate the unique contribution of each program over and above covariates. For comparing effect sizes across models, we utilized standardized coefficients (β), which were calculated using z-scored variables. The analyses were performed in R programming using the lm() function, taking into consideration fully observed cases with appropriate checks for multicollinearity and fit of the model.

Results

Descriptive statistics

The descriptive statistics for the analytic sample (n = 1,845) are presented in Table 1. Preceding the sample report, the same way the means of dummy variables are treated as proportions, and the medians with ranges are used for continuous variables; this sets up the regression by clarifying scale, typical values, and potential skew.

Table 1. Descriptive Statistics

Variable	Min	Median	Mean	Max
WICpreg	0.0	0.00	0.4010840	1.0

Variable	Min	Median	Mean	Max
AFDCpreg	0.0	0.00	0.1626016	1.0
bthwht	0.0	0.00	0.3512195	1.0
HOME97	7.9	20.60	20.2493767	27.0
AGE97	3.0	7.00	7.4487805	13.0
faminc97	0.0	43,153.58	53,558.6398157	784,610.6
readss97	47.5	101.50	102.5005420	165.5

Reading scores (readss97) have a distribution of 47.5 to 165.5, with a median of 101.5, mean of 102.5, and standard deviation of 16.0, which means there is a centered distribution relatively to age norms. Age ranges in children (3-13 years of age, median 7, mean 7.45), therefore the achievement can be observed from early childhood to early adolescence, where we hope age and home environment will show appreciable signal for achievement. Family income (2002 USD) shows the expected right-skew (min 0, median $43,154, mean $53,559, max $784,611). This skew reinforces the decision to control for income in levels (as in the assignment), while outliers of income greater than 1 million may have an effect on the margin of their income as well. The HOME97 index clusters around the low 20s (range 7.9-27.0, median 20.6, mean 20.25) with a moderate variation in home cognitive–emotional stimulation; based on the aforementioned theory, a positive correlation with reading ability is expected to be observed.

Regression Results

We estimated four nested OLS models to assess how prenatal WIC and AFDC participation relate to children's reading achievement after accounting for age, family income, low birth weight, and the home environment, shown in Table 2. The controls-only model (1) explains 17.8% of the variance in reading scores. Adding WIC (model 2) raises R^2 to 0.188 ($\Delta R^2 \approx 0.010$), and adding AFDC instead (model 3) raises R^2 to 0.185 ($\Delta R^2 \approx 0.007$). Including both

policy indicators (model 4) yields $R^2 = 0.191$, a net gain of ($\Delta R^2 \approx 0.013$) over controls. For all models tested, F-statistics are highly significant, and residual standard errors decreasing slightly with the addition of predictors indicates improvements in fit.

All covariates are expectedly associated when the model is fully specified (model 4). The coefficient for age is positive, b $\approx$ 0.43, with reading scores of older children being comparatively higher. There is a small but statistically significant positive association as well for income (b = 0.00005, *), that is, the coefficient implies +0.5 points for each approximate $10,000 increase, and about +1 point for income coefficient of +1 each in the larger models. The statistical significance (b $\approx$ 1.18–1.40, * across models) supports an assertion that you could say the following example: HOME97 difference of 5-point corresponds to around 6-7 points on the reading score. The association between low birth weight and the ability to read is also significant and as expected (b ≈ -2.2, *), mirroring the universal underpinnings of developmental deficiencies.

If we look at the measures of policy in regard to birth weight, all indicators are negative and statistically significant with controls included. WIC in model 2 is -3.82,* and it decreases to -3.12* in the full model; AFDC in model 3 is -3.81* and it decreases to -2.40 to reach the full model ($p < .05$). The diminishing effect of this model $\frac{2}{3}$ in comparison with model 4 (self-helped) shows the tendency for these participants having common problems and issues. In reality, the negative coefficients here are not due to the harm of the program; what happens is that the program attracted families with greater risk related to the socioeconomic and health issues, therefore the indicators are likely catching the negative selection as well as the remaining unobserved disadvantage of the families already controlled by birth weight, income, and home

environment. The R^2 pattern shows that WIC adds data variance slightly more than AFDC, with the values of 0.010 and 0.007 beyond the controls. The two programs seem to explain a small but still meaningful portion of the variance while correlating with the stronger covariates. Still, WIC has a somewhat greater impact on this population. Finally, the results remind that family and home factors almost make up the scale for differences in achievement, while program factors are only picked up largely as risk indicators for enrolled families rather than causing the impacts of these programs.

Table 2. Regression Results

	Dependent variable:			
	readss97			
	(1)	(2)	(3)	(4)
AGE97	0.440***	0.424***	0.449***	0.432***
	(0.122)	(0.122)	(0.122)	(0.122)
faminc97	0.0001***	0.00005***	0.0001***	0.00005***
	(0.00001)	(0.00001)	(0.00001)	(0.00001)
bthwht	-2.531***	-2.277***	-2.341***	-2.203***
	(0.738)	(0.736)	(0.737)	(0.735)
HOME97	1.398***	1.219***	1.281***	1.177***
	(0.122)	(0.127)	(0.125)	(0.128)
WICpreg		-3.819***		-3.123***
		(0.783)		(0.839)
AFDCpreg			-3.806***	-2.397**
			(0.978)	(1.045)

Constant	68.787***	74.466***	71.858***	75.365***
	(2.377)	(2.633)	(2.496)	(2.659)
Observations	1,845	1,845	1,845	1,845
R^2	0.178	0.188	0.185	0.191
Adjusted R^2	0.176	0.186	0.182	0.188
Residual Std. Error	14.525 (df = 1840)	14.435 (df = 1839)	14.469 (df = 1839)	14.419 (df = 1838)
F Statistic	99.465*** (df = 4; 1840)	85.318*** (df = 5; 1839)	83.214*** (df = 5; 1839)	72.140*** (df = 6; 1838)
Note:	*p<0.1; **p<0.05; ***p<0.01			

Conclusion

This article analyzes the effect of prenatal exposure to WIC and AFDC programs participation on children's reading skills in later age. Family income, age, birth weight, and home environment are also considered. The controls alone explained a meaningful share of variance, and adding program indicators produced small but non-trivial gains in fit (R^2 rising from 0.178 to 0.191). Only three variables - age, family income, and HOME97- remained significant in the full model, with positive relationships among all of them, while low birth weight was significantly negatively associated with reading, which was consistent with both developmental input and the environment implications. In contrast, the coefficients for WIC and AFDC were significantly negative. Thus, we concluded that they were harmful to the family, but rather those who enrolled were poorly selected families, especially at the observable and unobservable dimensions of disadvantage. Even after employing controls, the residual disadvantage present in the enrolled families could not be ruled out. These could be considered not as program harms, but results of adverse selection: families who register are, by design, more disadvantaged and, namely, those who are disadvantaged on both observed and unobserved dimensions. The overall

pattern involves WIC data explaining slightly more unique variance than AFDC, although both effects appear minimal in comparison with the family and home factors.

Some limitations should be considered when making causal inferences. This is an observational study containing complete-case data, so it is not able to fully control for omitted variables (including parental education, prenatal health, and neighborhood context) and measurement issues (skew of income, potential nonlinearity, and timing mismatches). Another area of future research should involve the testing of robustness of different specifications (log income, robust standard errors, propensity weighting, or sensitivity analyses) and, where applicable, the use of instrumental variables, regression discontinuity, and parallel thresholds to control selection effects and separate program effects. Policy implications, though, point toward bundled supports: prenatal nutrition (WIC) and income assistance (AFDC, TANF-positive) should be sustained, but paired with home-learning and early literacy interventions, given that the strongest associations were found with family resources and quality of home environment.

References

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Bradley, R. H., Caldwell, B. M., Rock, S. L., Hamrick, H. M., & Harris, P. (1988). Home observation for measurement of the environment: Development of a home inventory for use with families having children 6 to 10 years old. *Contemporary Educational Psychology, 13*(1), 58–71. https://doi.org/10.1016/0361-476X(88)90006-9

Brooks-Gunn, J., & Duncan, G. J. (1997). The effects of poverty on children. *The Future of Children, 7*(2), 55–71. https://doi.org/10.2307/1602387

Appendix: R code

```
good <- read.csv('/Users/leeeileen/Downloads/good.csv')
#descriptive stats
df <- na.omit(good[, c("WICpreg", "AFDCpreg", "readss97", "AGE97", "faminc97
", "bthwht", "HOME97")])
summary(df)
desc <- data.frame(
  Variable = c("WICpreg", "AFDCpreg", "bthwht", "HOME97", "AGE97", "faminc97
", "readss97"),
  Min = sapply(df[c("WICpreg", "AFDCpreg", "bthwht", "HOME97", "AGE97", "fami
nc97", "readss97")], min, na.rm=TRUE),
  Median = sapply(df[c("WICpreg", "AFDCpreg", "bthwht", "HOME97", "AGE97", "f
aminc97", "readss97")], median, na.rm=TRUE),
  Mean = sapply(df[c("WICpreg", "AFDCpreg", "bthwht", "HOME97", "AGE97", "fam
inc97", "readss97")], mean, na.rm=TRUE),
  Max = sapply(df[c("WICpreg", "AFDCpreg", "bthwht", "HOME97", "AGE97", "fami
nc97", "readss97")], max, na.rm=TRUE)
)
ft <- flextable(desc)
ft <- autofit(ft)
ft <- fontsize(ft, size = 10)
save_as_docx(ft, path = "descriptive.docx")
# Regression Models
# Model 0: Controls only
m0 <- lm(readss97 ~ AGE97 + faminc97 + bthwht + HOME97, data = df)
# Model 1: Add WIC
m1 <- lm(readss97 ~ AGE97 + faminc97 + bthwht + HOME97 + WICpreg, data = df)
# Model 2: Add AFDC
m2 <- lm(readss97 ~ AGE97 + faminc97 + bthwht + HOME97 + AFDCpreg, data = df)
# Model 3: Add both WIC + AFDC
m3 <- lm(readss97 ~ AGE97 + faminc97 + bthwht + HOME97 + WICpreg + AFDCpreg,
data = df)
stargazer(m0, m1, m2, m3, type = "html", out = "models.html")
rmarkdown::pandoc_convert("models.html", to = "docx", output = "models.docx")
#Results, compare R-squared values
summary(m0)$r.squared
summary(m1)$r.squared
summary(m2)$r.squared
summary(m3)$r.squared
#Calculate changes in R-squared
delta_wic <- summary(m1)$r.squared - summary(m0)$r.squared
delta_afdc <- summary(m2)$r.squared - summary(m0)$r.squared
delta_both <- summary(m3)$r.squared - summary(m0)$r.squared
cat("delta R² WIC:", delta_wic, "\n")
cat("delta R² AFDC:", delta_afdc, "\n")
cat("delta R² Both:", delta_both, "\n")
```